

# TRAN VAN NGOC

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## EDUCATION

### Cao Thang Technical College

Information Technology (Web Development Major)

Ho Chi Minh City

Aug 2024 – Present

- **Cumulative GPA:** 7.9/10.0 (Year 2 GPA: 7.93/10.0).
- **Specialization:** Focused on Modern Web Development (React, Next.js, Node.js), Database Design (PostgreSQL), Object-Oriented Programming (OOP), and software development lifecycle.
- **Academic Standing:** Currently in the 3rd year (5th semester), preparing for the graduation project and industry internship.

## PERSONAL PROJECTS

### Hyundai Ecommerce B2B Platform

Mar 2026 – Present

GitHub | Live Demo

Next.js 16, PostgreSQL, Drizzle ORM, Turborepo

- **Architecture & Type-Safety:** Architected a typesafe **Turborepo monorepo** using Next.js 16, React 19, and Drizzle ORM to enforce unified compile-time validation from database schemas to client-side views.
- **Database & Transactions:** Designed PostgreSQL schemas to support multi-warehouse inventory and customer-tiered pricing, implementing database transactions to guarantee data consistency during concurrent checkouts.
- **Edge Performance:** Optimized storefront loading speeds using Next.js 16 Cache Components ("use cache", cacheLife, cacheTag) to cache dynamic requests at the edge and minimize database overhead.
- **API Security:** Secured server endpoints by building typesafe Server Actions validated with custom **Zod schemas** to eliminate input validation bypasses and prevent malicious payloads.
- **CI/CD & Verification:** Automated test suites and continuous integration (CI) pipelines using **GitHub Actions** and Bun Test to verify system stability prior to release.

### High-Concurrency Ticket Booking System

May 2026 – Jun 2026

GitHub | Live Demo

NestJS, PostgreSQL, Redis, Docker, Cloudflare

- **Concurrency Control:** Resolved ticket overselling and seat booking race conditions under high concurrent load by implementing database-level **Pessimistic Locking** (FOR UPDATE) and Redis Distributed Locks (Redlock).
- **Async Architecture:** Decoupled heavy invoice processing and notification tasks from the request-response cycle using **BullMQ asynchronous workers** and synchronized real-time state with WebSockets (Socket.io).
- **Secured DevOps:** Integrated a zero-downtime **Blue-Green deployment** pipeline via Caddy Server routed through **Cloudflare Tunnel**, locking down the host by exposing **zero public inbound ports** (22/80/443).
- **Resource Optimization:** Mitigated Out-Of-Memory (OOM) crash risks on a low-resource 1GB RAM VM by reducing system RAM consumption by **35%** (727MB to 480MB) using Docker systemd overrides (250MB limit), swap tuning, and service pruning.

## SKILLS

**Back-End Technologies:** NestJS, Node.js, REST APIs, WebSockets (Socket.io), Redis, Session-based Auth, Drizzle ORM.

**Front-End Technologies:** Next.js 16, React 19, TypeScript, Tailwind CSS, shadcn/ui, Zustand.

**Databases & DevOps:** PostgreSQL, Docker, Docker Compose, Git, Turborepo, Bun, Doppler, GitHub Actions CI/CD.

**Languages:** English (Professional document reading and comprehension).

## PERSONAL STRENGTHS

**Strong Discipline:** Consistently adhere to regulations and processes during work.

**Responsibility & Dedication:** Ensure quality and timely delivery of assigned tasks. Responsible use of AI.

## INTERESTS

Physical exercise (Calisthenics), exploring emerging technologies.